

Advanced C++ Bank Management System

Object-Oriented Programming (OOP) Implementation - Eng_Zeyad

Project Specifications:

- Language: C++ (OOP Principles: Encapsulation, Classes)
- Features: Account Creation, Deposit, Withdraw, Balance Inquiry
- Status: Clean, production-ready code with interactive CLI interface.

Source Code (bank_system.cpp)

```
#include <iostream>
#include <vector>
#include <string>
#include <iomanip>

using namespace std;

class BankAccount {
private:
    int accountNumber;
    string accountHolder;
    double balance;

public:
    BankAccount(int accNo, string holder, double initialBalance) {
        accountNumber = accNo;
        accountHolder = holder;
        balance = initialBalance;
    }

    int getAccountNumber() const {
        return accountNumber;
    }

    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            cout << "
[✓] Successfully deposited: $" << fixed << setprecision(2) << amount;
        } else {
            cout << "
[✗] Invalid deposit amount!";
        }
    }

    void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            cout << "
[✓] Successfully withdrew: $" << fixed << setprecision(2) << amount;
        } else {
            cout << "
[✗] Insufficient balance or invalid amount!";
        }
    }

    void displayAccountInfo() const {
```

```

        cout << "
        -----";

        cout << "
Account Number : " << accountNumber;
        cout << "
Account Holder : " << accountHolder;
        cout << "
Current Balance : $" << fixed << setprecision(2) << balance;
        cout << "
        -----";

    }
};

int main() {
    vector<BankAccount> accounts;
    int choice;

    do {
        cout << "

=== ADVANCED BANK MANAGEMENT SYSTEM ===";
        cout << "
1. Create Account";
        cout << "
2. Deposit Money";
        cout << "
3. Withdraw Money";
        cout << "
4. View Account Details";
        cout << "
5. Exit";
        cout << "
Enter your choice (1-5): ";
        cin >> choice;

        switch (choice) {
            case 1: {
                int accNo;
                string name;
                double initBal;
                cout << "Enter Account Number: ";
                cin >> accNo;
                cout << "Enter Holder Name: ";
                cin >> name;
                cout << "Enter Initial Balance: ";
                cin >> initBal;
                accounts.push_back(BankAccount(accNo, name, initBal));
                cout << "

[✓] Account created successfully!";
                break;
            }

            case 2: {
                int accNo;
                double amount;
                cout << "Enter Account Number: ";
                cin >> accNo;
                cout << "Enter Amount to Deposit: ";
                cin >> amount;
                bool found = false;
                for (auto &acc : accounts) {
                    if (acc.getAccountNumber() == accNo) {
                        acc.deposit(amount);

```

```

        found = true;
        break;
    }
}
if (!found) cout << "[x] Account not found!";
break;
}
case 3: {
    int accNo;
    double amount;
    cout << "Enter Account Number: ";
    cin >> accNo;
    cout << "Enter Amount to Withdraw: ";
    cin >> amount;
    bool found = false;
    for (auto &acc : accounts) {
        if (acc.getAccountNumber() == accNo) {
            acc.withdraw(amount);
            found = true;
            break;
        }
    }
    if (!found) cout << "[x] Account not found!";
    break;
}
case 4: {
    int accNo;
    cout << "Enter Account Number: ";
    cin >> accNo;
    bool found = false;
    for (const auto &acc : accounts) {
        if (acc.getAccountNumber() == accNo) {
            acc.displayAccountInfo();
            found = true;
            break;
        }
    }
    if (!found) cout << "[x] Account not found!";
    break;
}
case 5:
    cout << "
Thank you for using the system. Goodbye!
";
    break;
default:
    cout << "[x] Invalid choice, please try again.";
}
} while (choice != 5);

return 0;
}

```